

Demonstrating Pinching Visuo-haptic Display: A Dynamic Cloth Texturization Via Visually Enhanced Tactile Sensation

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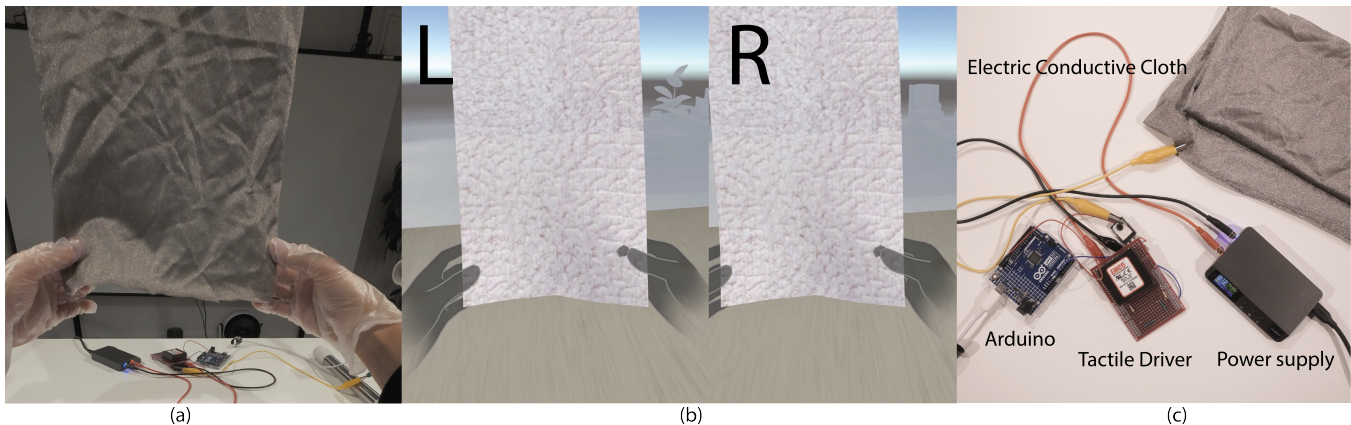


Figure 1: Overview of our study. Our study demonstrates how visual information can modulate the tactile perception of cloth. A cloth is placed in VR space in the same position and size as the cloth in physical space, but with a different texture. Its texture can be changed by the administrator. (a) Users touch the cloth with and without the VR HMD. Users can experience different tactile sensations with and without it - Users perceive that if it looks rough, it is rough to touch. Users can try a variety of tactile condition of the physical cloth with only one cloth, because the cloth is Pinching Tactile Display, a cloth that changes tactile sensation by electrostatic adsorption [5]. (b) The cloth in the VR environment mirrors the movement of the cloth in physical space. It deforms when touched. In this way, users do not lose immersion when touching the VR cloth. (c) The administrator can change the tactile sensation of the physical cloth by controlling the electricity applied to the cloth. In our implementation, we can control the tactile sensation by controlling the voltage and frequency of the square wave applied to the cloth.

Abstract

Fabric is integral to everyday life, from clothing to furniture, and its tactile qualities greatly influence comfort. Although digital reproduction of fabric textures promises to broaden access to personalized textiles, achieving realistic haptic feedback remains challenging. This study investigates how visual overlays affect tactile perception in a Pinching Tactile Display (PTD). We augmented the physical feedback of an electrically conductive cloth (ECC) with synchronized virtual textures in a virtual reality environment. User

evaluations revealed that visual cues significantly modified perceived fabric sensations, including roughness and thickness. These findings highlight the potential of cross-modal integration for more immersive fabric representation, offering guidelines for future haptic design in applications such as online fashion retail and VR-based interaction.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Mixed / augmented reality*; • **Hardware** → **Emerging interfaces**.

Keywords

Haptic Display, Textiles, Soft Actuators, Personal Fabrication

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1 Introduction

Fabric is a material people interact with on a daily basis. People wear fabric clothing every day. Many pieces of furniture in our daily spaces use fabrics, such as sofas, pillows, and chairs. We feel the tactile sensation of them every time we touch it with our hands or brush it against our skin. Its tactile sensation is soft and pleasant to touch, so it is an indispensable material for people to live a comfortable life.

If these tactile qualities align well with individual preferences, the overall daily experience becomes more pleasant. If the fine nuances of the tactile sensation of the fabric can be conveyed by a computer, it will be possible to transcend the limitations of time and space that humans have and confirm the tactile sensation of the fabric. Then we would be able to confirm the tactile sensation of the fabric at any time and from any place. This will allow us to take the time to look for clothes that feel good to the touch from the comfort of our own homes, without having to go directly to a clothing store. This would be especially valuable in online shopping for the fashion industry, where subtle tactile differences that are difficult to convey through images and words alone could be communicated to the consumer [4]. In addition, in the fashion industry, where virtual reality applications for the fashion industry [12] are rapidly advancing in the marketplace, haptic feedback for fabric textures will enhance the inherent immersive appeal of VR and bring us closer to realizing interactions with clothing, such as virtual try-ons [7]. In this way, the ability to present fabric textures in these virtual spaces would expand the scope and richness of online shopping experiences. However, displaying the nuances of the tactile sensation of cloth is a challenge. Fabric is soft, flexible, and has subtle surface textures that cannot be easily captured or reproduced with conventional haptic hardware [1, 2, 10].

Several systems have been developed to provide tactile experiences with fabric. Haptic Revolver [9] and Haptic Palette [3] display physical cloth to display tactile sensation of cloth. In addition, Telexiles proposed a contrastive learning model to match similar tactile cloth. However, these have physical space requirements that limit the variety of cloth it can display, there were limitations in conveying the delicate nuances of cloth. Another haptic approach, "Pinching Tactile Display" (PTD) [5], uses an electrically conductive cloth (ECC) to emulate different fabric sensations by manipulating voltages and frequencies. PTD maintains the softness of the fabric during interaction, and can continuously control its roughness, giving it an advantage in displaying nuances of tactile sensation of the fabric. However, it only changes the roughness of a cloth and its expressiveness remains limited, highlighting the need for improved expressiveness and more refined control mechanisms to faithfully reproduce subtle cloth properties.

An idea from the field of multisensory perception is that tactile experiences are strongly influenced by visual cues. When users see a surface that appears soft or rough, their brains integrate these visual signals with tactile inputs, either enhancing or altering the perceived sensation [7]. Consequently, pairing the immersive visuals of VR with devices such as PTD can potentially overcome

the limitations of the other and have the possibility to convey more nuanced and convincing fabric representations, resulting in an expanded range of perceived textures. This investigation could help bridge small perceptual gaps that might arise in PTD-based fabric rendering.

Consequently, the purpose of this study is to investigate how the visual overlay of textures affects the tactile sensation provided by the PTD. We conducted a user study in which participants interact with a Pinching Tactile Display augmented by virtual textures. By aligning the physical feedback of the ECC with carefully rendered VR visuals, we explored how visual cues enhance tactile perception for cloth.

Our contributions can be summarized as follows.

- Proposing a visually enhanced dynamic cloth texturization system using VR: We overlay target fabric visuals in a virtual environment onto a physical ECC device, synchronizing the movement of the physical cloth with the on-screen texture.
- Conducting a user study to test our hypothesis on tactile perception: We evaluated whether overlaying different visual textures (via VR) affects the perceived tactile sensation when using Pinching Tactile Display (PTD).

2 Proposed Method

To investigate how the visual overlay of texture affects the tactile experience provided by the PTD, we developed a pinching visual-haptic display: dynamic cloth texturization via visually enhanced haptic perception. We developed a cloth that adjusts the position and size of the physical cloth in VR space. The movement of the cloth is made resemble the physical cloth using simulation, and when the user touches the virtual cloth, the cloth visually moves with the user's hand movement. This allows the user to touch the physical cloth without losing the immersive experience of the cloth's visuals in VR space. This allows the user to confirm the tactile sensation of the cloth, as if it were a cloth whose texture only changes, but whose actual texture and tactile properties remain the same. Because the cloth is an established cloth in VR space, the administrator can place different textures on the cloth without changing its size or position. This overcomes the physical limitation of the difficulty of preparing a cloth where only the texture is changed but the tactile characteristics remain the same. This allowed us to investigate how the texture change affects the tactile sensation when the cloth is touched.

As a physical cloth, we used PTD, a cloth whose tactile properties are changed by electrostatic adsorption. This cloth is made of conductive fabric, and by controlling the pattern of electricity on the cloth, the tactile sensation when a person touches the cloth can be changed. Therefore, by controlling the displayed tactile sensation, users can try several physical cloth tactile sensation properties with a single piece of cloth. Thus, with only one setup, the user can feel the influence of visual cues on tactile perception in different cloth tactile states.

2.1 VR Modality

To make the virtual cloth resemble the physical cloth, we performed real-world tracking, simulated the cloth motion, and matched the size of the virtual cloth to the physical cloth. For real-world tracking,

Texture	Roughness	Stiffness	Thickness	Warmth
denim	0.147	0.902	0.561	0.743
gauze	0.0671	0.459	0.862	0.685
towelling	0.00547*	0.183	0.371	0.696
voile	0.645	0.148	0.272	0.156

Table 1: ANOVA results for Texture in VR effects on tactile perceptions. * shows significant results. It shows that by seeing textures through VR, for each texture, for roughness, stiffness, thickness, and warmth, showing $Pr(>F)$

Term	$Pr(>F)$
Volt	0.00154*
Freq	0.0972
Texture	0.00547*
Volt $\times$ Freq	0.00423*
Volt $\times$ Texture	0.0367*
Freq $\times$ Texture	0.330
Volt $\times$ Freq $\times$ Texture	0.711

Table 2: The effect of texture of towelling on tactility. * is a significant result. In the context of texture, the interaction between voltage and texture is significant.

we integrate real-time tracking information from the Meta Quest Pro controller mounted above the ECC. We attached the Meta Quest Pro controller to the ECC and sent position data to Unity. For cloth motion simulation, we ran spring-mass simulations and high-resolution texture rendering on the cloth using Unity. The textures we implemented are denim, gauze, toweling, and voile. This kept the virtual cloth aligned with the real position of the ECC when the user manipulated or moved the cloth.

Cloth Simulation We employed in Unity a spring-mass system [6] to simulate the behavior of the cloth. Each mesh vertex is treated as a mass point, and the edges are modeled as springs that enforce structural, shear, and bending constraints. Parameters such as spring stiffness and damping coefficients are tuned to approximate four representative fabric types: denim, gauze, toweling, and voile, as shown in Figure ?? . We used high-resolution materials from Quixel Megascan to enhance realism in the VR environment. The visually rendered cloth in VR thus responds dynamically to user interactions or movements, simulating real cloth deformations.

World Blending To track the real world, we mounted the Meta Quest Pro controller on a support structure (hanger) above the ECC. In Unity, the virtual cloth object is parented to the controller, ensuring that the real world movements of the ECC translate one-to-one to the virtual cloth. We also replicated the physical environment in Unity by measuring and modeling reference objects (tables, floors) at a one-to-one scale. This calibration ensures that when the user’s hand approaches the visually rendered cloth, it corresponds exactly to the location of the real ECC. Because of this careful spatial calibration, users perceive both the real and virtual cloth as overlapping, significantly reducing the mismatch between visual and haptic feedback.

2.2 Tactile Feedback Modality

To provide the tactile sensation of fabric, we utilize PTD to convey various aspects of fabric touch. By altering the voltage and frequency applied to the ECC, we can modulate its perceived texture. The PTD in our setup is centered on applying electrostatic forces to a conductive cloth. By varying voltage and frequency inputs, the system can adjust the roughness that users feel when pinching the cloth. We use a conductive cloth of approximately 25 cm $\times$ 45 cm, made of 100% silver fiber with an electrical resistance of about 1 Ω . An Arduino UNO generates square waves at 50 Hz, 100 Hz, and 200 Hz within 0–5 V. A MOS-FET (TK31N60X) switches these signals to deliver 100–300 V to the cloth, effectively creating variable electrostatic forces. Users wear ultra-thin polyvinyl chloride (PVC) gloves (0.035 mm thick) to ensure safe interactions while still perceiving tactile differences. For safety measure, We did a protective enclosure for high-voltage components, current-limiting resistors to guard against unintended electrical surges, use of insulating PVC gloves to minimize direct exposure to high voltage. These measures allow participants to safely pinch and explore the cloth, focusing on the perceived changes in tactile sensation.

3 User Study

To investigate how the visual overlay of textures affects the tactile sensation provided by the PTD, we conducted a user study. We experimented how much the tactile sensation changes when we touch the cloth with and without VR, i.e., with and without texture. In addition, to examine how the effect of appearance is affected by the tactile feel of the physical cloth, we varied the voltage and frequency of the pulse wave applied to the pinching tactile display and examined each of them. We hypothesized that the perceived tactile sensation of the fabric would be closer to the impression of the visual texture due to cross-modal completion by the brain.

3.1 Experimental Design

Four participants (three men and one woman) aged 25–30 participate in the experiment. Each participant touched the same conductive cloth under various conditions: in the VR condition and without VR (control). Without VR condition, participants directly saw the bare ECC (conductive cloth) without a VR head-mounted display (HMD), and in the VR-on condition, participants wore a VR HMD displaying visual textures such as denim, gauze, towelling, or voile depending on the trial. The voltage levels tested were 0V, 50V, 100V, 150V, 200V, 250V, and 300V, while the frequency levels were 50Hz, 100Hz, and 150Hz. This resulted in a total of 19 PTD conditions, calculated as six voltage levels multiplied by three frequency levels plus the 0V condition. To minimize order effects, the presentation order of these 19 states was randomized for each participant. Additionally, there were five texture conditions (four VR textures and one no VR condition), and the presentation order of these five states was also randomized for each participant. Consequently, the total number of states amounted to 95 (19 PTD conditions multiplied by five texture conditions).

For each state, participants touched the PTD and reported the tactile sensation using 1 – 5 likert scale. Of the tactile qualities of the fabrics, they answered about roughness, stiffness, thickness, and warmth. Our choice of descriptive term pairs relates to the findings

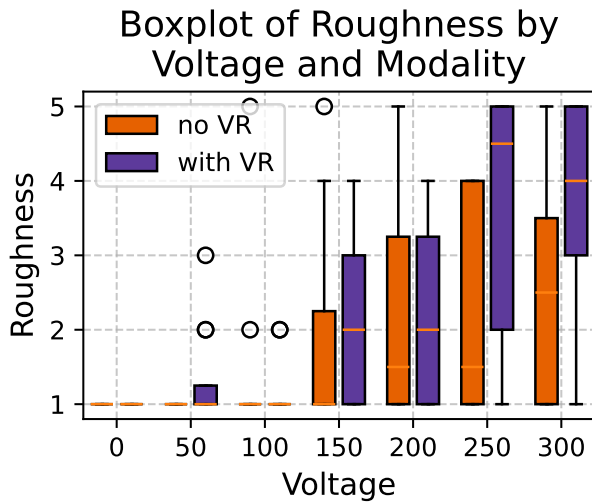


Figure 2: This graph shows the perception of roughness when participants touched the cloth while looking at a towel placed in a virtual reality (VR) room and when they touched the cloth without looking at the VR room. The results show that participants tended to perceive the roughness more when they touched the towel while looking at it than when they did not look at the VR room. This suggests that the tactile sensation of roughness may be enhanced when looking at a rough-textured cloth.

of Soufflet, Calonnier, and Dacremont [8] who demonstrate that the most significant scales their participants used to rate the properties of textiles were stiff-flexible, thick-thin, and soft-harsh.

3.2 Results and Discussion

Since our data is a three-way, within-participant design (voltage $\times$ frequency $\times$ texture) and the normality assumption may not always hold, we adopted the aligned rank transform (ART) followed by a three-way ANOVA [11]. Table 1 shows the ANOVA results comparing how each of the four textures affected the four tactile properties. We found that a significant effect was detected on *roughness* ($p = 0.00547$). On the other hand, for denim, gauze, voile, no statistically significant differences were observed for any properties, although gauze trended toward significance ($p = 0.0671$ for roughness). These results suggest that the visual display of texture with prominent weave improves participants' subjective assessment of roughness compared to the non-VR condition. On the contrary, denim and voile may be visually more similar to baseline ECC, resulting in no significant effect on tactile perception.

To examine toweling conditions more closely, we analyzed roughness ratings across different voltage and frequency levels on that condition. Table 2 and Fig. 2 showed the result. We observed a significant interaction (Voltage $\times$ VR) whereby higher voltages amplified the effect of visual roughness. In particular, at higher voltages (e.g. 300V), participants reported greater roughness in the VR condition than in the non-VR condition. In contrast, at lower voltages

(50–100V), both conditions yielded similar ratings, suggesting that roughness perception gets larger when the original physical cloth roughness was high, the visual effect gets larger.

For future work, investigating a broader spectrum of fabrics (e.g., corduroy, fleece, lace) to identify which visual-tactile parameters most significantly affect perception. This could reinforce the finding that the visual roughness of a texture influences tactile perception. Additionally, it may allow us to determine whether the visual roughness is more affected by the height of the fabric's weave or by the fineness of the weave. From another perspective, introducing temperature cues to augment cross-modal feedback could further enhance the realism of fabric simulation and might reveal how multiple sensory channels interact to form unified tactile impressions.

4 Conclusion

In this study, we propose a pinching visual-haptic display: dynamic cloth texturization via visually enhanced haptic perception. We investigated how the visual overlay of textures affects the tactile sensation with this system. The results are that toweling display increased participants' subjective ratings of roughness compared to non-VR conditions, suggesting that the visual display of texture with prominent weave improves participants' perception of roughness. This result can be used to modulate fine roughness perception for cloth.

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References

- [1] Carissa J Cascio and K Sathian. 2001. Temporal cues contribute to tactile perception of roughness. *Journal of Neuroscience* 21, 14 (2001), 5289–5296.
- [2] Seungmoon Choi and Katherine J Kuchenbecker. 2012. Vibrotactile display: Perception, technology, and applications. *Proc. IEEE* 101, 9 (2012), 2093–2104.
- [3] Donald Degraen, Anna Reindl, Akhmadjon Makhsadov, André Zenner, and Antonio Krüger. 2020. Envisioning Haptic Design for Immersive Virtual Environments. In *Companion Publication of the 2020 ACM Designing Interactive Systems Conference (Eindhoven, Netherlands) (DIS'20 Companion)*. Association for Computing Machinery, New York, NY, USA, 287–291. doi:10.1145/3393914.3395870
- [4] Dhruv Grewal, Scott Motyka, and Michael Levy. 2018. The Evolution and Future of Retailing and Retailing Education. *Journal of Marketing Education* 40 (02 2018), 027347531875583. doi:10.1177/0273475318755838
- [5] Takekazu Kitagishi, Hirotaka Hiraki, Hiromi Nakamura, Yoshio Ishiguro, and Jun Rekimoto. 2024. Pinching Tactile Display: A Cloth that Changes Tactile Sensation by Electrostatic Adsorption. In *Proceedings of the 2024 International Conference on Advanced Visual Interfaces (Arenzano, Genoa, Italy) (AVI '24)*. Association for Computing Machinery, New York, NY, USA, Article 26, 9 pages. doi:10.1145/3656650.3656690
- [6] Tiantian Liu, Adam W. Bargteil, James F. O'Brien, and Ladislav Kavan. 2013. Fast simulation of mass-spring systems. *ACM Trans. Graph.* 32, 6, Article 214 (nov 2013), 7 pages. doi:10.1145/2508363.2508406
- [7] Marina Ricci. 2022. Exploiting Virtual Reality for Enhancing the Shopping Experience in the Fashion Industry: Between Interaction and Perception. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. 938–941. doi:10.1109/ISMAR-Adjunct57072.2022.00210
- [8] Yvanne Soufflet, Maurice Calonnier, and Catherine Dacremont. 2004. A comparison between industrial experts' and novices' haptic perceptual organization: A tool to identify descriptors of the handle of fabrics. *Food quality and preference* 15, 7–8 (2004), 689–699.
- [9] Eric Whitmire, Hrvoje Benko, Christian Holz, Eyal Ofek, and Mike Sinclair. 2018. Haptic Revolver: Touch, Shear, Texture, and Shape Rendering on a Reconfigurable Virtual Reality Controller. In *Proceedings of the 2018 CHI Conference on Human*

- Factors in Computing Systems* (Montreal QC, Canada) (*CHI '18*). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3173574.3173660
- [10] Michael Wiertlewski. 2011. *Reproduction of tactual textures: transducers, mechanics, and signal encoding*. Ph. D. Dissertation. Université Pierre et Marie Curie-Paris VI.
- [11] Jacob O. Wobbrock, Leah Findlater, Darren Gergle, and James J. Higgins. 2011. The aligned rank transform for nonparametric factorial analyses using only anova procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (*CHI '11*). Association for Computing Machinery, New York, NY, USA, 143–146. doi:10.1145/1978942.1978963
- [12] Huiyue Wu, Yu Wang, Jiali Qiu, Jiayi Liu, and Xiaolong Zhang. 2018. User-defined gesture interaction for immersive VR shopping applications. *Behaviour & Information Technology* 38 (12 2018), 1–16. doi:10.1080/0144929X.2018.1552313